Closed-Loop Remote Sensing Image Change Detection via Active Difference-Prior Guidance

Jingyu Yang, *Member, IEEE*, Jiayang Li, Chengqi Bao, Jiyuan Li, Jianwu Dang, and Jiaxin Li, *Member, IEEE*

***Abstract*—Deep learning-based methods have achieved remarkable progress in urban change detection (CD) from high-resolution remote sensing (RS) images. However, most existing methods treat different information merely as a static output, without allowing the statistical and semantic priors embedded in it to continuously guide subsequent inference. As a result, they remain limited in both pseudo-change suppression and multiscale change representation. To address these issues, this article proposes a difference-prior-guided network (DPGNet). First, a statistical prior-guided rectification module (SPRM) is developed to derive priors from the statistical characteristics of bitemporal features, which are then used to jointly compensate for imaging-style discrepancies, thereby yielding more discriminative difference representations from the outset. Second, a difference retrospective probe (DRP) is designed to treat preliminary difference features as hypothesized priors and revisit the original spatiotemporal context to verify whether the detected changes are genuine, thereby further refining change semantics. Finally, a semantic prior-guided aggregation module (SPGA) exploits big difference features as global semantic priors to guide shallow detail features toward semantic alignment, thus preserving the structural integrity of large-scale targets while enhancing sensitivity to subtle changes. On the LEVIR-CD, WHU-CD, and SYSU-CD datasets, DPGNet achieves $F1$-scores of 92.35%, 94.49%, and 84.16%, respectively, outperforming existing state-of-the-art (SOTA) methods and validating its effectiveness for urban CD from high-resolution RS images. Meanwhile, DPGNet contains only 7.75-M parameters and 8.66-G floating-point operations (FLOPs), showing a favorable balance between accuracy and efficiency. In addition, DPGNet still has room for improvement in detecting change regions with ambiguous boundaries and extremely small and spatially scattered change targets. The source code is publicly available at: https://github.com/AspirinRS/DPGNet**

***Index Terms*—Change detection (CD), contextual retrospection, difference-prior guidance, semantic-guided aggregation.**

I. Introduction

REMOTE sensing (RS) image change detection (CD) aims to identify surface changes in the same geographic area at different times and plays a key role in tasks such as environmental monitoring [1], [2], land management [3], [4], and disaster assessment [5], [6]. With the increasing scene complexity of high-resolution RS images, the problems of intraclass spectral variation and interclass spectral similarity have become more prominent. Although differences between bitemporal images remain key cues for detecting land surface changes, complex appearance variations can no longer be directly associated with real changes. Difference responses between bitemporal images are not only easily affected by external factors such as illumination variation and seasonal differences, which disturb the statistical distribution in feature space and introduce pseudo changes, but are also often unable to fully characterize real changes due to complex target structures and varying scales (as shown in Fig. 1).

First, cross-temporal imaging variations can cause significant appearance differences for the same object. As shown in Fig. 1(a), under strong illumination changes, the same building exhibits clear differences in bitemporal images, and the resulting interference can easily cause false detections in unchanged regions. Second, Fig. 1(b) shows that the complex geometric structures and heterogeneous textures of buildings can lead to discontinuous local feature responses, resulting in structural holes and fragmented boundaries. Finally, as shown in Fig. 1(c), in large and complex scenes, the weak responses of small targets are easily submerged in background clutter, which often results in missed detections.

Traditional CD methods mainly fall into three categories: algebra-based, transformation-based, and classification-based approaches. Algebra-based methods directly compare pixel differences between bitemporal images through arithmetic operations, such as image differencing [7], image ratioing [8], and image quantization [9], and then generate change maps using thresholding or clustering. Transformation-based methods, such as principal component analysis (PCA) [10] and Kauth–Thomas (KT) transformation [11], map features into specific spaces to enhance the separability of change information. Classification-based methods assign semantic labels to each pixel using classifiers, including random forests [12] and support vector machine (SVM) [13], and infer change regions by comparing classification results from different times. However, due to their sensitivity to noise and radiometric inconsistency, these methods struggle to handle

Received 27 March 2026; revised 29 June 2026; accepted 15 July 2026. Date of publication 20 July 2026; date of current version 28 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62367005, in part by the Natural Science Foundation of Gansu Province under Grant 24JRRA253 and Grant 24JRRA236, in part by the Central Guidance Project for Local Science and Technology Development under Grant 24ZYQA051, in part by the Science and Technology Research Program of Chongqing Municipal Education Commission under Grant KJQN202620606, in part by the Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation under Grant GZB20260443, and in part by the Chongqing Postdoctoral Science Foundation under Grant 2025CQBSHTB2012. *(Corresponding authors: Jingyu Yang; Jiayang Li; Jiaxin Li.)*

Jingyu Yang, Jiayang Li, Jiyuan Li, and Jianwu Dang are with the School of Electronic and Information Engineering, Lanzhou Jiaotong University, Lanzhou 730070, China (e-mail: yangjy@mail.lzjtu.cn; lijiayang320@163.com; 3302059540@qq.com; dangjw@mail.lzjtu.cn).

Chengqi Bao is with the School of Automation and Electrical Engineering, Lanzhou Jiaotong University, Lanzhou 730070, China (e-mail: 610653720@qq.com).

Jiaxin Li is with the School of Computer Science and Technology (National Exemplary Software School), Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: lijiaxin203@mails.ucas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3714892

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

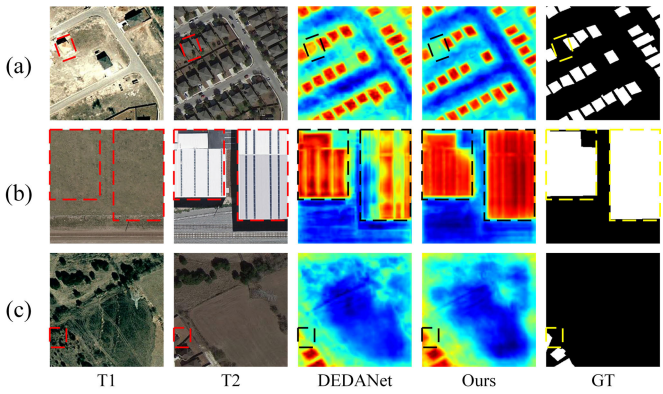


Fig. 1. Three typical challenges in CD. (a) Spectral differences caused by inconsistent imaging conditions lead to false alarms in unchanged regions. (b) Complex geometric structures of objects can cause discontinuous structural difference responses, resulting in structural holes in large-scale targets. (c) Small-scale targets produce only weak difference features and are therefore prone to missed detection.

high-resolution RS images with rich details and complex changes.

In recent years, deep neural networks (DNNs) [14], [15], [16] have been widely applied to CD. Since Caye Daudt et al. [17] introduced fully convolutional networks (FCNs) [18] into this task, convolutional neural network (CNN)-based methods have become a major research direction in CD. To improve the representation of change features, researchers have developed various enhanced architectures. For example, Fang et al. [19] used dense skip connections to preserve localization information in deep features, while Huang et al. [20] adopted reparameterized large-kernel convolutions to enlarge the effective receptive field and enhance feature extraction in complex scenes. However, constrained by the local inductive bias of convolution, these methods still struggle to model long-range dependencies. To overcome this bottleneck, vision transformer-based methods [21] have emerged. Among them, Bandara and Patel [22] employed a hierarchical Transformer to efficiently capture multiscale long-range details, while Wang et al. [23] further combined global and local attention to compensate for the limitations of single-scale feature modeling. However, although Transformer architectures enhance global modeling, their self-attention mechanisms remain susceptible to irrelevant background interference in complex scenes, which amplifies redundant information and weakens the focus and representation of real change regions.

Deep learning-based CD generally involves four stages: bitemporal feature extraction, difference feature generation, multiscale fusion, and change classification. In existing studies, some methods enhance feature representation by designing more complex backbones, such as deep CNNs, Transformers, or Mamba. However, relying solely on backbone enhancement often fails to provide sustained performance gains. We argue that the current bottleneck lies not only in feature extraction, but also in how differences are modeled from the extracted features. Existing methods often lack in-depth modeling of the crucial difference generation stage, preventing high-quality features from being transformed into discriminative difference representations. At this stage, most methods remain constrained by fixed feature measurement paradigms, such as simple channel concatenation or algebraic differencing, and treat difference features merely as passive terminal outputs. As a result, they fail to exploit key information such as cross-temporal statistical shifts and therefore cannot actively correct appearance shifts caused by inconsistent imaging conditions. Furthermore, the initially generated difference features inherently contain valuable change cues and spatial priors, yet existing one-way inference schemes lack a mechanism to convert them into high-confidence dynamic priors and actively reuse them for retrospective verification. This issue further extends to the multiscale fusion stage, where the lack of guidance from high-level difference semantics makes simple linear fusion insufficient to coordinate cross-level semantics, causing subtle changes to be submerged and large-scale objects to become fragmented.

To address the limitations of existing methods, this article proposes a difference-prior-guided network (DPGNet). Unlike traditional one-way inference schemes, we argue that robust CD should be a progressive confidence-refinement process, in which change cues inferred at one stage are transformed into guiding priors for the next stage and further used to constrain relevant feature representations, thereby enabling difference verification and feature calibration. Specifically, the bitemporal images are first fed into a weight-sharing lightweight backbone, MobileNetV2, to extract multilevel features. To overcome the limitations of conventional static measurement methods, we design a statistical prior-guided rectification module (SPRM), which uses statistical priors of features to actively correct imaging-style discrepancies and generate more discriminative difference features at the source. On this basis, a difference retrospective probe (DRP) further builds an adaptive feedback mechanism by using the preliminary differences as probes to retrospectively query the original feature space for authenticity verification, thereby effectively removing pseudo changes. In addition, a semantic prior-guided aggregation module (SPGA) uses deep difference priors to calibrate shallow details in a top-down manner. By bridging multiscale semantic misalignment, it enables precise activation of subtle changes while preserving the structural integrity of large-scale targets. Finally, the network outputs a high-precision binary change map. The main contributions of our work are summarized below.

1) We propose DPGNet for urban CD from high-resolution RS images. It transforms different cues inferred at earlier stages into guiding priors for later stages, thereby constructing a feature-level closed-loop guidance framework for difference verification and feature calibration within a single forward pass.
2) We design a difference modeling mechanism that integrates statistical calibration and retrospective verification. Specifically, SPRM uses statistical priors to actively correct imaging-style deviations, while DRP introduces an adaptive retrospective verification process that converts preliminary differences into probes to retrospectively query the original feature space, thereby

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

effectively suppressing false changes caused by environmental interference.

3) We propose SPGA, which uses big differences as explicit semantic priors to dynamically select effective shallow details for enhancing small-target localization, while preserving the structural integrity of large-scale change regions through high-level semantic continuity.
4) Experiments on three challenging high-resolution datasets demonstrate that the proposed DPGNet outperforms state-of-the-art (SOTA) methods in detection accuracy while remaining computationally competitive.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the proposed network and its details. Section IV reports the experimental results and analysis. Section V further discusses the effectiveness and design rationality of the proposed method. Section VI concludes this article.

II. Related Work

With the rapid development of deep learning for CD, different representation enhancement, multiscale feature fusion, and guidance mechanism design have gradually become important research directions. This section reviews related work from these three perspectives.

A. Difference Feature Enhancement-Based CD Methods

Difference feature generation and enhancement aim to extract more discriminative change information from bitemporal features and are crucial for suppressing background interference and improving real-change localization. Early deep learning methods mostly adopted siamese architectures [24]. Typical examples are FC-Siam-conc and FC-Siam-diff proposed by Daudt et al. [17], which generate difference features through channel concatenation and absolute differencing, respectively. However, differencing is sensitive to cross-temporal noise and appearance variations, while concatenation often fails to highlight discriminative differences. To address these limitations, later studies introduced multiscale convolutions and attention mechanisms to improve the scale adaptability and discriminative ability of different features. For example, STANet [25] employed spatiotemporal attention to improve robustness to illumination changes and registration errors, while HANet [26] used hierarchical attention with parallel convolutions and row-column attention for fine-grained difference feature selection. To enhance spatiotemporal consistency, Li et al. [27] exploited spatiotemporal memory to mine attention cues for stronger change representation, and Pan et al. [28] used spatial weighting to improve temporal consistency and suppress background noise. Huang et al. [29] further introduced coordinate attention to enhance spatiotemporal difference representation and improve the modeling of global and local change information. Recent studies enhance difference representation through long-range dependency modeling and frequency information mining. HAM-CD [30] uses Mamba and hybrid attention to model local and global relationships, while OSAWaveNet [31] improves change discrimination through semantic clustering and frequency-enhanced difference learning.

In addition, to improve robustness in difference computation, some studies have moved from simple differencing to similarity-based measures. For example, Dong et al. [32] replaced direct subtraction with cosine similarity at the channel and spatial levels to characterize feature differences. Fu et al. [33] further introduced structural similarity index measure (SSIM), replacing pixel-wise comparison with structural differences within local windows to better preserve structural consistency. However, such methods usually do not explicitly model global distribution shifts caused by variations in imaging conditions. Without statistical calibration, they cannot effectively distinguish imaging-style variations from real target changes.

B. Multiscale Difference Feature Fusion-Based CD Methods

To balance deep semantics and shallow details, multiscale feature aggregation is widely used in CD. A common strategy aligns feature resolutions across levels before fusion. For example, Fang et al. [34] use an MLP decoder to upsample and concatenate multiscale features, while GeoFormer [35] improves change-region matching via dense upsampling. However, dense upsampling alone can hardly bridge the cross-level semantic gap, and repeated interpolation may blur boundary details. To mitigate this, feature reweighting has been introduced to emphasize key scales and improve discriminability. For instance, Lu et al. [36] proposed MFPF-Net with a feature redistribution module for adaptive multiscale fusion, and Luo et al. [37] combined skip connections with reweighting to extract discriminative spatiotemporal features. Nevertheless, reweighting alone often fails to suppress background redundancy, causing subtle change signals to be overwhelmed during cross-scale fusion.

To address the shortcomings of simple fusion, recent studies have investigated fine-grained interaction and progressive aggregation of multiscale features to exploit interlevel complementarity. For example, Huang et al. [38] constructed a progressive decoder with dynamic weights to integrate high-frequency details and low-frequency context layer by layer, while Xing et al. [39] employed multireceptive-field convolutions to progressively enhance multiscale target localization. Wang et al. [40] established multilevel feature dependencies in both channel and spatial dimensions through a hybrid attention module. Li et al. [41] combined Mamba with frequency-domain convolutions to capture deep dependencies and enable progressive multiscale decoding. Although these strategies alleviate scale differences to some extent, explicit modeling of interlevel semantic gaps remains insufficient. Without deep semantic guidance, shallow noise cannot be effectively suppressed, causing small targets to be submerged in the background and large-scale change regions to lose structural integrity.

C. Feature Guidance-Based CD Methods

Recent feature-guided modeling has attracted attention in RS tasks [42], [43]. In CD, explicit prior features [44], [45] are also used to compensate for the susceptibility of implicit features to background noise, helping focus on key regions and rectify feature bias. In terms of structural guidance, existing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

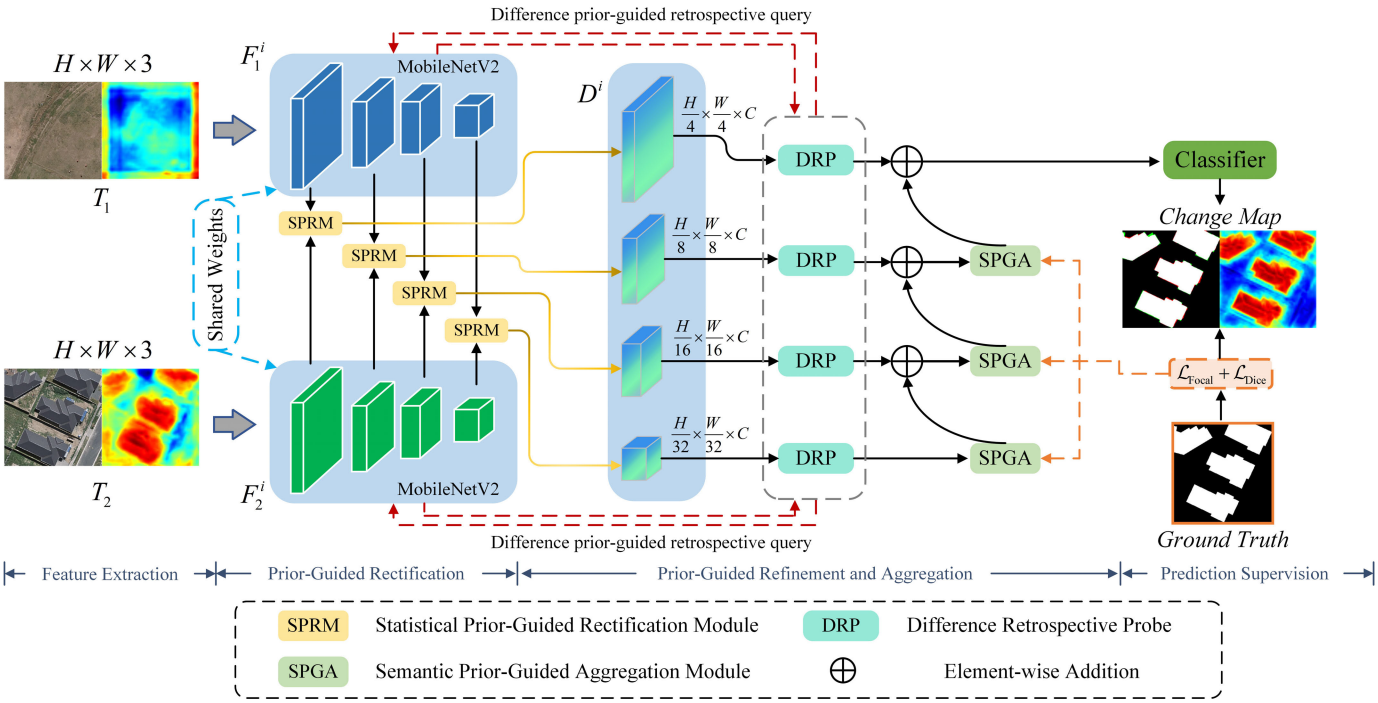


Fig. 2. Overall architecture of DPGNet. A weight-sharing MobileNetV2 extracts multilevel features from bitemporal images. SPRM calibrates imaging deviations using statistical priors and generates difference representations; DRP retrospectively queries the bitemporal context to verify change authenticity; SPGA uses deep semantics to guide shallow-detail selection and aggregation. Finally, a binary change map is produced.

methods often exploit edge priors to strengthen boundary constraints. For example, Chen et al. [46] used an edge-aware map to explicitly guide the decoder, sharpening boundary details while suppressing background noise. EFICNN [47] combined a Laplacian pyramid with reverse attention to guide the network in jointly capturing high-frequency edges and target body regions. EdgeRefNet [48] improves boundary localization through edge-guided feature refinement. Such structured priors have shown good performance in alleviating boundary blurring and region adhesion. In contrast, some studies focus on using spatiotemporal evolution or mask information to guide feature optimization. For example, Wang et al. [49] used a spatiotemporal mask to guide feature fusion in decoding and compensate for information loss, while Xu et al. [50] dynamically adjusted parameters in a Mamba architecture based on temporal and mask semantics to improve scene adaptability. FarmCD [45] targets farmland CD by generating farmland masks in a weakly supervised manner and employing a gating mechanism to enhance farmland-related change features. To focus more directly on change targets, other studies introduce coarse change maps as priors. For example, Song and Jiang [51] appended a learned spatial attention map to deep features to directly guide the network in separating change targets from complex backgrounds. Wei et al. [52] generated a guidance map by fusing shallow and deep features, and injected it unidirectionally into low-level features to correct semantic bias.

Although existing studies have introduced various external priors for guidance, such exogenous signals are constrained by the reliability of prior estimation and thus cannot stably align with real change regions in complex scenes. Some studies have attempted to use endogenous differences as guidance, but this is often limited to one-shot injection at a single stage or level, making it difficult to suppress the cascading amplification of initial spurious differences or adapt to the dynamic demands of multilevel representation transitions. In particular, change cues from small targets are easily diluted by background responses during progressive convolution and fusion, causing terminal or single-point guidance to fail due to the premature loss of critical information. In summary, it is necessary to explore a feature-level closed-loop guidance mechanism that spans the entire process of difference modeling, verification, and fusion. Such a mechanism can transform different cues from modeled objects into priors that guide subsequent verification and fusion within a single forward pass, thereby further alleviating the performance bottleneck of CD in complex scenarios.

III. Methodology

This section first presents an overview of the proposed DPGNet architecture, and then introduces its three core modules, namely, SPRM, DRP, and SPGA. Finally, the adopted hybrid loss function is described.

A. Overall Architecture

As shown in Fig. 2, the proposed DPGNet adopts a lightweight, weight-sharing MobileNetV2 backbone to extract multilevel features from bitemporal images $T_1, T_2 \in \mathbb{R}^{H\times W\times 3}$, yielding $\{F_1^i\}_{i=1}^4$ and $\{F_2^i\}_{i=1}^4$. These features are then fed into SPRM, which calibrates imaging-style deviations using statistical priors and produces more discriminative multiscale difference representations $\{D^i\}_{i=1}^4$, thereby suppressing pseudo

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

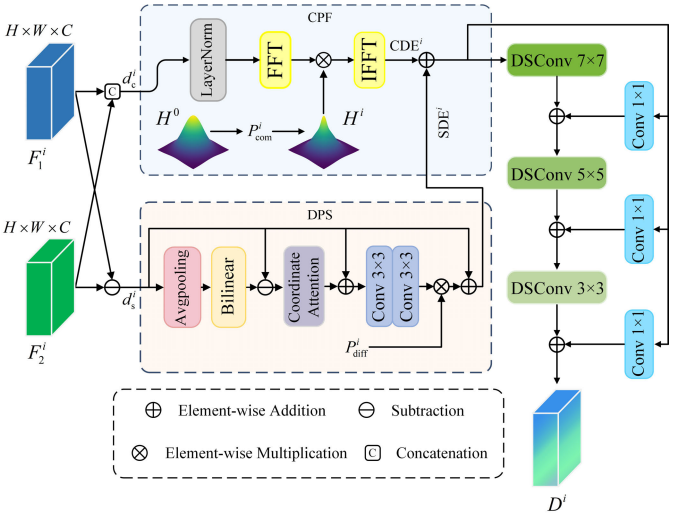


Fig. 3. Illustration of SPRM. CPF denotes the commonality statistical prior-guided frequency-domain filtering branch, and DPS denotes the discrepancy prior-guided spatial structure optimization branch.

changes caused by environmental interference at the source. Based on this, DRP treats the different representations as probes and retrospectively queries the bitemporal context to verify change authenticity. The verified features are further input to SPGA, where a top-down semantic-prior feedback pathway guides shallow-detail selection and multiscale aggregation to alleviate semantic misalignment during fusion. Finally, the prediction head outputs a binary change map. In this process, the three modules collaborate through progressive modeling of the difference prior: SPRM generates a candidate difference prior to be verified, DRP verifies it using bitemporal contextual evidence, and SPGA feeds the verified deep difference semantics back into shallow aggregation. Through this process, the different information is gradually transformed from an initial candidate response into a verified guiding prior, thereby constraining subsequent feature calibration and change discrimination.

B. Statistical Prior-Guided Rectification Module

To address the susceptibility of existing CD methods to style drift and complex-scene interference, we design a SPRM. It constructs two complementary dynamic priors, namely, commonality and discrepancy priors, from bitemporal feature statistics. The commonality statistical prior calibrates the global frequency-domain background, while the discrepancy statistical prior optimizes spatial structural responses, yielding more robust difference representations.

As shown in Fig. 3, SPRM first constructs the channel-wise difference feature d_c^i and the spatial difference feature d_s^i in parallel from the bitemporal features

$$\begin{aligned} d_c^i &= \mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Concat}\left(F_1^i, F_2^i\right)\right)\right) \\ d_s^i &= \left|F_1^i - F_2^i\right| \end{aligned} \tag{1}$$

where Concat(·) denotes concatenation along the channel dimension, and |·| denotes the absolute value operation.

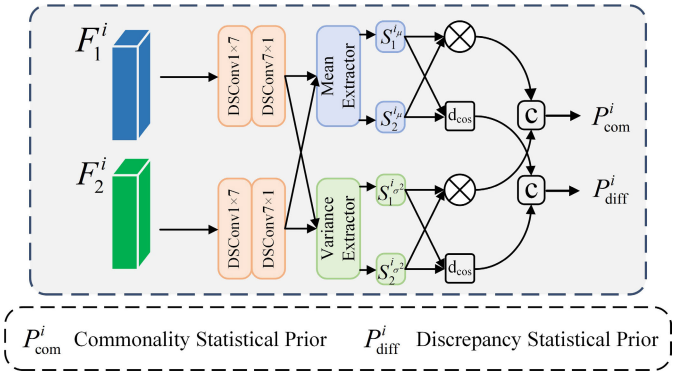


Fig. 4. Illustration of the statistical prior generation process, where d_{cos} denotes a discrepancy measure constructed based on cosine relationships.

1) Commonality Statistical Prior-Guided Frequency-Domain Filtering Branch: When processing the initial channel-wise difference features d_c^i, global filtering mechanisms [53] can be used for frequency-domain reweighting and denoising to reduce pseudo changes in complex scenes. However, such methods usually rely on static shared parameters, making it difficult to adapt frequency responses to the common background statistics of the current bitemporal inputs. To address this issue, we design a frequency-domain filtering branch guided by the commonality statistical prior.

This branch first applies LayerNorm to stabilize the feature distribution. Then, an FFT-based global filtering mechanism is introduced to project the features into the frequency domain for reweighting. The commonality statistical prior is used as a conditional signal to dynamically modulate the base frequency response, thereby enabling global frequency-domain calibration under background-consistency constraints. Inspired by CSI-Net [54], we construct it using the joint statistics of mean and variance. Specifically, as shown in Fig. 4, parallel 1×7 and 7×1 depthwise separable convolutions are first used for directional receptive-field modeling to reduce irrelevant background interference. Then, the mean and variance statistics of the bitemporal features are computed along the spatial dimension to extract first- and second-order statistical information, which can be formulated as follows:

$$\begin{aligned} S_t^{i,\mu} &= \mathrm{Mean}\left(\mathrm{StripDSConv}\left(F_t^i\right)\right), \qquad t = 1,2 \\ S_t^{i,\sigma^2} &= \mathrm{Var}\left(\mathrm{StripDSConv}\left(F_t^i\right)\right), \qquad t = 1,2 \end{aligned} \tag{2}$$

where StripDSConv denotes a directional convolution composed of parallel 1×7 and 7×1 depthwise separable convolutions. Mean(·) and Var(·) denote the mean and variance statistics computed along the spatial dimension, respectively.

Since the channel-wise difference features generated by concatenation mix the imaging-style variations of the two temporal inputs, this branch models the intersection of their statistics through element-wise multiplication to obtain the commonality statistical prior P_{com}^i, thereby emphasizing the shared background distribution and suppressing inconsistent fluctuations. This process is formulated as follows:

$$P_{\mathrm{com}}^i = \mathrm{Concat}\left(\phi\left(S_1^{i,\mu}\right)\otimes\phi\left(S_2^{i,\mu}\right), S_1^{i,\sigma^2}\otimes S_2^{i,\sigma^2}\right) \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

where $\otimes$ represents element-wise multiplication, and $\phi(\cdot)$ is a non-negative mapping function used to avoid sign ambiguity when negative mean values participate in the element-wise multiplication.

Finally, the generated P^i_{com} is used as a conditioning signal to dynamically modulate the base frequency response. Specifically, P^i_{com} is first mapped by a lightweight parameter network to produce a filter modulation increment ΔH^i. A prior modulation factor $1 + \alpha \cdot \tanh(\Delta H^i)$ is then constructed and applied to the base learnable filter H^0 through element-wise scaling, yielding the prior-modulated frequency-domain filter H^i. This process is formulated as follows:

$$\begin{aligned} \Delta H^i &= g_\theta\left(P^i_{\text{com}}\right) \\ H^i &= H^0 \times \left(1 + \alpha \cdot \tanh\left(\Delta H^i\right)\right) \end{aligned} \tag{4}$$

where $g_\theta(\cdot)$ denotes the parameter mapping function, α is the modulation strength coefficient, and $\tanh(\cdot)$ is used to constrain the modulation amplitude and avoid instability caused by overly strong perturbations.

In this process, P^i_{com} provides a background-consistency constraint for frequency-domain filtering, suppressing irrelevant frequency responses affected by imaging-style variations while preserving structurally discriminative difference components. The calibrated frequency response is subsequently mapped back to the spatial domain through a 2-D inverse fast Fourier transform (IFFT), yielding a more discriminative channel-wise difference encoding feature CDE^i. This process can be expressed as follows:

$$\text{CDE}^i = \text{IFFT}\left(\text{FFT}\left(d^i_{\text{c}}\right) \otimes H^i\right). \tag{5}$$

2) Discrepancy Statistical Prior-Guided Structure Optimization Branch: Although subtraction-generated spatial difference features can directly represent potential change regions, they often lead to broken responses and discontinuous boundaries. To address this issue, we design a discrepancy prior-guided structure optimization branch to enhance the spatial representation of real change regions from two aspects: local structure recovery and statistical discrepancy constraints.

To alleviate the smoothing effect of d^i_{s} at boundaries, we first apply average pooling to d^i_{s} to extract local low-frequency background components, and then restore them to the original resolution by bilinear interpolation. The difference between the original feature d^i_{s} and the pooled feature is then computed to suppress low-frequency smoothed background components and highlight the high-frequency contour information of change regions. Furthermore, the extracted edge feature d^i_{edge} is fed into a coordinate attention module, which integrates and recalibrates features along the horizontal and vertical directions, thereby preserving clear edge responses while enhancing structural consistency within target regions. The above process is formulated as follows:

$$\begin{aligned} d^i_{\text{edge}} &= d^i_{\text{s}} - f_{\text{bilin}}\left(f_{\text{ap}}\left(d^i_{\text{s}}\right)\right) \\ d^i_{\text{CoordAtt}} &= \text{CoordAtt}\left(d^i_{\text{edge}}\right) \oplus d^i_{\text{s}} \end{aligned} \tag{6}$$

where $f_{\text{ap}}(\cdot)$ denotes average pooling, and $f_{\text{bilin}}(\cdot)$ denotes bilinear interpolation. $\text{CoordAtt}(\cdot)$ denotes the coordinate attention operation, and $\oplus$ denotes element-wise addition. d^i_{CoordAtt} represents the spatial feature optimized by coordinate attention.

To further suppress unreliable local activations, we derive the discrepancy statistical prior P^i_{diff} from the statistical differences between the bitemporal features using mean and variance. Specifically, the cosine similarity between the two statistical descriptors is computed and then converted into a normalized angular discrepancy by the arccosine function, so as to characterize the inconsistency of the bitemporal statistical distributions. This process is defined as follows:

$$\begin{aligned} P^i_{\text{diff}} = \text{Concat}\Bigg(&\frac{1}{\pi}\arccos\left(\frac{\left\langle \phi\left(S_1^{i,\mu}\right), \phi\left(S_2^{i,\mu}\right)\right\rangle}{\left\|\phi\left(S_1^{i,\mu}\right)\right\|_2 \left\|\phi\left(S_2^{i,\mu}\right)\right\|_2 + \varepsilon}\right) \\ &\frac{1}{\pi}\arccos\left(\frac{\left\langle S_1^{i,\sigma^2}, S_2^{i,\sigma^2}\right\rangle}{\left\|S_1^{i,\sigma^2}\right\|_2 \left\|S_2^{i,\sigma^2}\right\|_2 + \varepsilon}\right)\Bigg) \end{aligned} \tag{7}$$

where $\langle\cdot,\cdot\rangle$ denotes the vector dot product, $\|\cdot\|_2$ denotes the L_2 norm, and ε is a stabilizing constant used to avoid a zero denominator.

This prior does not directly encode structural information; instead, it acts as a conditioning signal to modulate the preceding structural responses, thereby enhancing reliable activations, suppressing unreliable ones, and improving the structural discriminability of change regions. The modulated responses are then residually fused with d^i_{s}, preserving the original discrepancy cues and fine details while further strengthening structural representations. Finally, the spatial difference encoding feature SDE^i obtained under discrepancy-prior guidance is formulated as follows:

$$\text{SDE}^i = \left(\text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(d^i_{\text{CoordAtt}}\right)\right) \otimes P^i_{\text{diff}}\right) \oplus d^i_{\text{s}}. \tag{8}$$

After obtaining CDE^i and SDE^i, we first perform an initial fusion by element-wise addition. Subsequently, the features are refined by progressive depthwise separable convolutions with kernels from large to small, together with a 1×1 convolution for channel compression and information integration. Finally, the progressively refined difference feature D^i is obtained, which can be expressed as follows:

$$\begin{aligned} D^i_0 &= \text{CDE}^i \oplus \text{SDE}^i \\ D^i_1 &= \text{DSConv}_{7\times7}\left(D^i_0\right) \oplus \text{Conv}_{1\times1}\left(D^i_0\right) \\ D^i_2 &= \text{DSConv}_{5\times5}\left(D^i_1\right) \oplus \text{Conv}_{1\times1}\left(D^i_0\right) \\ D^i &= \text{DSConv}_{3\times3}\left(D^i_2\right) \oplus \text{Conv}_{1\times1}\left(D^i_0\right). \end{aligned} \tag{9}$$

C. Difference Retrospective Probe

Most existing CD methods follow a one-way inference pipeline, where difference features are generated from bitemporal features and then directly decoded into the final change map. In this process, unreliable local responses may be preserved and accumulated during subsequent propagation, while genuine change cues may be weakened or even submerged due to the lack of effective verification, thereby leading to both pseudo changes and missed detections. To this end, we design a DRP. Compared with existing interaction modeling methods that mainly focus on difference enhancement [55],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

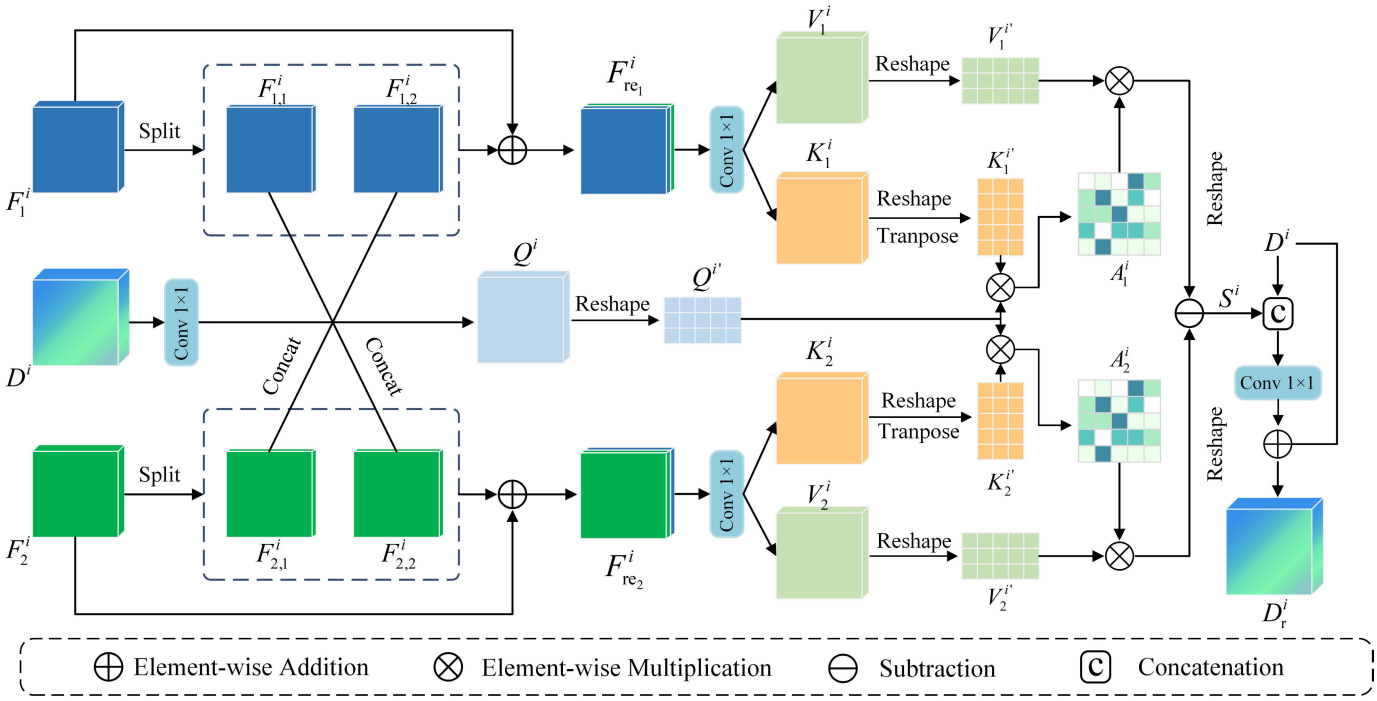


Fig. 5. Illustration of DRP.

DRP instead redefines the initial differences as change priors to be verified rather than final outputs, and further verifies and refines them by retrospectively querying the bitemporal context.

As shown in Fig. 5, to further enhance the fine-grained retrospective capability of different priors, we draw on the cross-reorganization idea of SSCFNet [56]. Specifically, F_1^i and F_2^i are first split into two equal-sized subfeature blocks along the channel dimension. The split subfeatures are then cross-interacted and reorganized to generate two mixed verification representations. Furthermore, the reorganized features are lightly projected by a 1×1 convolution, and residual connections are introduced to preserve the original basic context of each temporal input, thereby constructing two fine-grained verification representations. The formulation is given below

$$\begin{aligned} F_{\mathrm{re}_1}^i &= \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(F_{1,1}^i, F_{2,2}^i\right)\right) \oplus F_1^i \\ F_{\mathrm{re}_2}^i &= \mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(F_{1,2}^i, F_{2,1}^i\right)\right) \oplus F_2^i \end{aligned} \tag{10}$$

where $F_{1,1}^i$, $F_{1,2}^i$ and $F_{2,1}^i$, $F_{2,2}^i$ are the subfeature blocks obtained by splitting the bitemporal features F_1^i and F_2^i, respectively. $F_{\mathrm{re}_1}^i$ and $F_{\mathrm{re}_2}^i$ denote the two fine-grained reorganization-based verification representations.

Subsequently, $F_{\mathrm{re}_1}^i$ and $F_{\mathrm{re}_2}^i$ are linearly projected by a 1×1 convolution to generate the keys and values required for retrospective matching. Specifically, the keys K_1^i and K_2^i are used to model the matching relationship between the difference prior and the bitemporal context, while the values V_1^i and V_2^i carry the contextual information in the two verification representations. To facilitate subsequent attention-based matching, the key and value representations are further reshaped into 2-D tensors

$$K_t^i = \mathrm{Conv}_{1\times1}^K\left(F_{\mathrm{re}_t}^i\right), V_t^i = \mathrm{Conv}_{1\times1}^V\left(F_{\mathrm{re}_t}^i\right), t \in \{1,2\} \tag{11}$$

where $\mathrm{Conv}_{1\times1}^K(\cdot)$ and $\mathrm{Conv}_{1\times1}^V(\cdot)$ denote the 1×1 convolutional projections used to generate the key and value representations, respectively. They are then reshaped into the 2-D forms required for attention-based matching.

After constructing the two key-value branches, the difference feature D^i is further used to generate the query vector Q^i for active guidance of the subsequent retrospective matching. Unlike conventional self-attention, where the query, key, and value all originate from a common source feature, our method takes D^i as the query source and treats it as a difference prior to be verified, thereby explicitly initiating the matching and retrospective retrieval of bitemporal contextual evidence. The query Q^i is then matched with the two key branches K_1^i and K_2^i, respectively, to construct bitemporal retrospective attention weights under difference-prior guidance. The formulation is given as follows:

$$Q^i = \mathrm{Reshape}\left(\mathrm{Conv}_{1\times1}^Q\left(D^i\right)\right) \tag{12}$$

$$A_1^i = \mathrm{Softmax}\left(\frac{Q^i\left(K_1^i\right)^\top}{\sqrt{d_k}}\right), \quad A_2^i = \mathrm{Softmax}\left(\frac{Q^i\left(K_2^i\right)^\top}{\sqrt{d_k}}\right). \tag{13}$$

After obtaining the two attention weights A_1^i and A_2^i, they are applied to the value branches V_1^i and V_2^i, respectively, to extract the difference-supporting features R_1^i and R_2^i that match the difference feature D^i. This process performs cross-verification by retrospectively querying the underlying bitemporal features: if the initial difference prior is consistent with the low-level features, the corresponding response is enhanced; otherwise,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

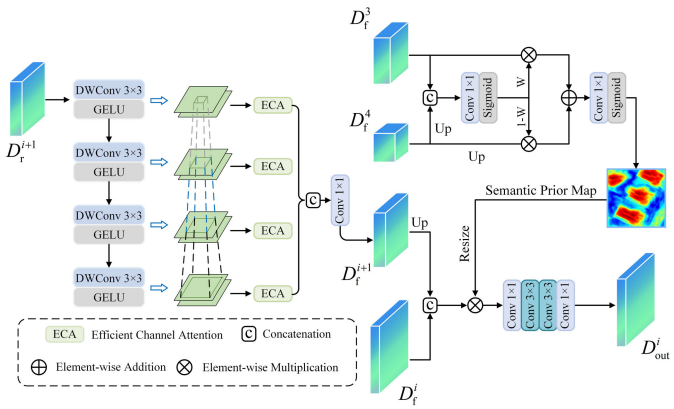


Fig. 6. Illustration of SPGA.

it is suppressed. The absolute difference between the two support features is then computed to capture the difference distribution S^i supported by the original context. Finally, S^i is concatenated with D^i, and a 1×1 convolution together with a residual connection is used for fusion, yielding the final retrospectively refined difference feature D_r^i. The formulation is given as follows:

$$R_t^i = \text{Reshape}\left(A_t^i \left(V_t^i\right)^\top\right), t \in \{1, 2\} \tag{14}$$

$$S^i = \left|R_1^i - R_2^i\right| \tag{15}$$

$$D_\text{r}^i = \text{Conv}_{1\times1}\left(\text{Concat}\left(D^i, S^i\right)\right) \oplus D^i. \tag{16}$$

D. Semantic Prior-Guided Aggregation Module

During decoding, effective aggregation of multiscale features is crucial for the complete recovery of change targets. However, most existing methods adopt FPN-like structures and fuse adjacent hierarchical features through simple concatenation or addition. Such static aggregation struggles to effectively handle the semantic gap across levels: shallow features preserve rich details but remain weak in semantics and are easily disturbed by noise, whereas deep features are semantically strong but lack fine spatial details. This inconsistency in cross-level feature representations makes it challenging for traditional static aggregation to simultaneously preserve the structural integrity of large-scale targets and achieve a precise perception of subtle changes.

To overcome this limitation, we propose the SPGA, which replaces traditional static multiscale fusion with explicit prior constraints to alleviate cross-level semantic misalignment and enhance the perception and localization of change targets at different scales. To further enrich the hierarchical contextual representation of different features and provide more reliable inputs for subsequent semantic-prior-guided aggregation, we first progressively enhance the input feature D_r^{i+1}. As shown on the left side of Fig. 6, D_r^{i+1} is progressively encoded by a stack of spatial depthwise convolutions with increasing dilation rates, so that its local texture responses gradually evolve into hierarchical representations with broader contextual coverage. Specifically, four consecutive 3×3 depthwise convolution layers are employed, with dilation rates set to $d_k \in \{1, 2, 4, 8\}$. For the kth feature, it can be formulated as follows:

$$D_k^{i+1} = \text{GeLU}\left(\text{DWConv}_{3\times3}^{d_k}\left(D_{k-1}^{i+1}\right)\right), \quad k = 1, 2, 3, 4 \tag{17}$$

where $D_0^{i+1} = D_\text{r}^{i+1}$ and $\text{DWConv}_{3\times3}^{d_k}$ denotes a 3×3 depthwise convolution with dilation rate d_k.

To suppress redundant noise and emphasize effective difference patterns at each level, efficient channel attention (ECA) is then introduced for lightweight recalibration, and the recalibrated feature is denoted as $\tilde{D}_k^{i+1}$. Finally, we merge the enhanced multilevel features along the channel dimension and use a 1×1 convolution to jointly perform hierarchical fusion and channel compression

$$D_\text{f}^{i+1} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\tilde{D}_1^{i+1}, \tilde{D}_2^{i+1}, \tilde{D}_3^{i+1}, \tilde{D}_4^{i+1}\right)\right). \tag{18}$$

On this basis, inspired by existing prior-guided multiscale fusion methods [57], we construct a semantic prior as an explicit guidance signal. Considering that a single-level feature often struggles to simultaneously preserve semantic discriminability and structural integrity, we do not derive the guidance directly from a single feature level. Instead, we jointly use the penultimate deep feature D_f^3 and the deepest feature D_f^4 as the prior source. Here, D_f^3 retains richer regional structures and boundary details, while D_f^4 provides stronger global semantic consistency. By jointly modeling them, a prior representation that captures both local structural information and high-level discriminative semantics can be obtained, thus providing more reliable semantic constraints for subsequent feature aggregation.

First, D_f^4 is upsampled to the same spatial resolution as D_f^3, and the two features are concatenated along the channel dimension to generate an adaptive weight map w, which adjusts their fusion ratio in semantic-prior construction. The weighted fusion of D_f^3 and the upsampled D_f^4 is then followed by a 1×1 convolution and a Sigmoid activation to produce the semantic prior map M_prior. This process preserves regional structural integrity while introducing more stable high-level semantic cues, thereby providing more reliable explicit constraints for subsequent feature aggregation. The formulation is given as follows:

$$w = \sigma\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(D_\text{f}^3, \text{Up}\left(D_\text{f}^4\right)\right)\right)\right) \tag{19}$$

$$D_\text{p} = w \otimes D_\text{f}^3 + (1 - w) \otimes \text{Up}\left(D_\text{f}^4\right) \tag{20}$$

$$M_\text{prior} = \sigma\left(\text{Conv}_{1\times1}\left(D_\text{p}\right)\right) \tag{21}$$

where $\text{Up}(\cdot)$ denotes bilinear interpolation upsampling, and σ denotes the Sigmoid activation function.

After obtaining the semantic prior map M_prior, we use it as an explicit guidance signal to constrain and guide cross-level feature aggregation during decoding. Specifically, as illustrated on the right side of Fig. 6, the hierarchically enhanced difference feature D_f^{i+1} is first upsampled to the same spatial resolution as the decoder feature D_f^i from the previous stage, and the two are merged along the channel dimension to form a candidate aggregation representation. The semantic prior, resized to the corresponding scale, is then applied to modulate the candidate feature in an element-wise manner, guiding features from different levels to be fused under a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

unified semantic constraint and improving the representational consistency of change targets at different scales. The process is described as follows:

$$D_{\text{agg}}^{i} = \text{Concat}\left(\text{Up}\left(D_{\text{f}}^{i+1}\right), D_{\text{f}}^{i}\right) \otimes M_{\text{prior}}. \tag{22}$$

Afterward, a 1×1 convolution is first used for channel compression, followed by two stacked 3×3 convolutions to locally integrate and refine the semantically constrained features. Finally, another 1×1 convolution produces the output difference feature D_{out}^{i} of the current layer. This process helps alleviate semantic misalignment in cross-level feature aggregation while improving the structural integrity and localization accuracy of change regions.

E. Hybrid Loss Function

In CD, changed regions usually occupy only a limited portion of the image, and complex backgrounds further aggravate the confusion between pseudo changes and real changes. To enhance the model's sensitivity to minority changed pixels and hard samples, while enhancing the overall consistency between predictions and real change regions, we optimize the network using a hybrid loss composed of focal loss and dice loss. Specifically, focal loss is adopted to alleviate class imbalance and is defined as follows:

$$\mathcal{L}_{\text{Focal}}\left(y_p, y_i\right) = -\alpha\left(1 - y_p\right)^{\gamma} \log\left(y_p\right). \tag{23}$$

Dice loss is introduced to constrain the overall overlap between the predicted region and the ground-truth region, and is defined as follows:

$$\mathcal{L}_{\text{Dice}}\left(y_p, y_i\right) = 1 - \frac{2y_i y_p + s}{y_i + y_p + s} \tag{24}$$

where $y_i \in \{0, 1\}$ represents the binary label of pixel i, and $y_p \in [0, 1]$ represents the predicted probability that the corresponding pixel belongs to the changed class. α is the weighting factor, γ is the adjustable focusing parameter, and s is a smoothing constant introduced to prevent division by zero.

Considering that the network produces multilevel difference representations during decoding, we further impose deep supervision on each output level to constrain the progressive optimization of difference representations. The total loss is thus defined as follows:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{4}\left(\lambda\mathcal{L}_{\text{Focal}}^{(k)} + \mathcal{L}_{\text{Dice}}^{(k)}\right) \tag{25}$$

where $k \in \{1, 2, 3, 4\}$ indexes the multiscale decoding stages, and λ is the balancing coefficient, which is set to 0.5 in this article.

IV. Experiments and Analysis

A. Datasets

Experiments are conducted on three public RS CD datasets for training, validation, and testing, namely, LEVIR-CD [25], WHU-CD [58], and SYSU-CD [59].

1) *LEVIR-CD:* This dataset consists of 637 pairs of bitemporal high-resolution RS images collected from Google Earth. The original image size is 1024×1024 pixels with a spatial resolution of 0.5 m. The dataset is mainly designed for building a CD. Following the official split, the images are cropped into nonoverlapping patches of 256×256, resulting in 7120 training samples, 1024 validation samples, and 2048 test samples.
2) *WHU-CD:* This dataset is mainly used for building CD and contains bitemporal high-resolution images of the Christchurch region in New Zealand, acquired before the 2011 earthquake and after reconstruction in 2016. The original image size is $32\,507 \times 15\,354$ pixels. The images are cropped into 256×256 patches, and 10% of the training set is randomly selected as the validation set, resulting in 5947 training samples, 743 validation samples, and 744 test samples.
3) *SYSU-CD:* This dataset consists of 20 000 pairs of high-resolution aerial image patches of 256×256 pixels, with a spatial resolution of 0.5 m, covering various change scenarios in Hong Kong, China. Following the official 6:2:2 split, 12 000 pairs are used for training, 4000 for validation, and 4000 for testing.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: In the experiments, the proposed DPGNet is implemented on the PyTorch platform and trained on an NVIDIA GeForce RTX 4090 GPU with 24 GB memory. The model is trained for 200 epochs with a batch size of 32. A cosine annealing schedule is adopted for the learning rate, with an initial value of 0.0005 and a 20-epoch warm-up stage. AdamW is used as the optimizer, with a weight decay of 0.0005 and a dropout rate of 0.1. In the hybrid loss, the focal loss parameters are set to $\alpha = 0.25$ and $\gamma = 4$, while the balancing coefficient λ is set to 0.5. To further improve the model's generalization capability, a unified data augmentation strategy was adopted during training, including flipping, cropping, random temporal swapping, and Gaussian blurring.

2) Evaluation Metrics: To quantitatively evaluate CD performance, five commonly used metrics are adopted in this article, including precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA), which are defined as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{26}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{27}$$

$$\text{F}_1 = 2 \cdot \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{28}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{29}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{TN} + \text{FP}} \tag{30}$$

where TPs, TNs, FPs, and FNs denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. All metrics range from 0 to 1, and higher values indicate better CD performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISON RESULTS ON THE LEVIR-CD, WHU-CD, AND SYSU-CD DATASETS. FOR EACH METRIC, THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY. ALL RESULTS ARE REPORTED IN PERCENTAGE (%)

Methods	LEVIR-CD					WHU-CD					SYSU-CD				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
FC-EF$_{2018}$	90.01	87.26	88.62	79.56	98.86	93.62	75.53	83.61	71.83	98.75	78.17	78.98	78.57	64.70	89.84
SNUNet$_{2021}$	89.55	87.80	88.67	79.64	98.86	66.30	91.32	76.82	62.36	97.68	86.91	60.11	71.07	55.12	88.46
BIT$_{2022}$	90.16	88.31	89.22	80.54	98.91	77.92	86.14	81.83	69.24	98.48	81.39	73.10	77.02	62.63	89.71
ChangeFormer$_{2022}$	92.01	88.81	90.40	82.48	99.04	89.44	76.37	82.39	70.04	98.71	81.59	73.28	77.21	62.88	89.80
TFI-GR$_{2022}$	92.15	89.73	90.93	83.36	99.09	94.08	91.65	92.85	86.66	99.44	83.76	78.25	80.91	67.94	91.29
BiFA$_{2024}$	91.94	89.26	90.58	82.79	99.06	95.43	92.86	94.13	88.91	99.51	86.98	81.44	84.12	72.59	92.75
SEIFNet$_{2024}$	91.96	88.09	89.98	81.79	99.00	86.16	90.28	88.17	78.84	99.04	83.36	78.67	80.95	67.99	91.27
STADE-CDNet$_{2024}$	92.47	87.69	90.02	81.84	99.01	91.50	79.62	85.14	74.12	98.90	74.50	76.18	75.33	60.42	88.23
DEDANet$_{2025}$	91.21	91.26	91.24	83.89	99.11	94.17	92.99	93.57	87.92	99.49	80.27	85.87	82.97	70.90	91.69
SFEARNet$_{2025}$	92.12	90.13	91.11	83.68	99.10	95.23	92.47	93.83	88.38	99.52	86.62	80.18	83.27	71.34	92.40
CDMamba$_{2025}$	91.23	90.93	91.08	83.62	99.11	95.38	92.75	94.05	88.76	99.53	82.79	83.45	83.12	71.12	92.01
ChangeTitans$_{2025}$	92.15	90.70	91.42	84.20	99.13	95.00	91.70	93.32	87.48	99.48	85.00	81.32	83.12	71.12	92.21
OSAWaveNet$_{2026}$	91.83	90.63	91.23	83.87	99.12	93.94	91.74	92.83	86.62	99.44	83.85	81.50	82.66	70.43	91.82
HAM-CD$_{2026}$	92.10	91.32	91.71	84.69	99.15	92.53	93.42	92.97	86.86	99.45	86.72	80.25	83.36	71.47	92.45
EdgeRefNet$_{2026}$	92.45	90.38	91.40	84.16	99.13	93.80	90.58	92.16	85.46	99.39	83.80	81.20	82.48	70.18	91.62
DPGNet(Ours)	92.50	92.20	92.35	85.80	99.22	95.49	93.51	94.49	89.55	99.57	86.90	81.59	84.16	72.64	92.76

C. Comparison With SOTA Methods

To verify the effectiveness of DPGNet, we compare it with 15 SOTA CD methods, including FC-EF [17], SNUNet [19], ChangeFormer [22], TFI-GR [60], BIT [61], BiFA [62], SEIFNet [29], STADE-CDNet [27], DEDANet [41], SFEARNet [63], CDMamba [64], ChangeTitans [65], OSAWaveNet [31], HAM-CD [30], and EdgeRefNet [48]. Among them, FC-EF and SNUNet are CNN-based methods, while ChangeFormer, BIT, and STADE-CDNet adopt Transformer architectures. CDMamba and HAM-CD introduce a state-space model for global modeling. ChangeTitans employs a neural memory mechanism to enhance long-range contextual modeling, while EdgeRefNet improves change boundary localization through edge-guided refinement. TFI-GR, STADE-CDNet, SEIFNet, DEDANet, SFEARNet, OSAWaveNet, and HAM-CD are more closely related to our method, as they all improve CD performance in complex scenes through difference-feature enhancement strategies. To ensure fairness, all comparison methods are retrained and evaluated based on the official model codes released by their authors, and are compared under unified experimental settings, including the same data split, input size, data augmentation, and evaluation metrics.

1) Quantitative Analysis: Table I reports the quantitative comparison results of different methods on the LEVIR-CD, WHU-CD, and SYSU-CD datasets. Overall, DPGNet achieves the best results on most evaluation metrics across the three datasets, demonstrating clear performance advantages.

On the LEVIR-CD dataset, DPGNet achieves the best $F1$-score of 92.35% and IoU of 85.80%. Compared with recent methods such as ChangeTitans, OSAWaveNet, HAM-CD, and EdgeRefNet, DPGNet still maintains superior overall performance, indicating that our method has stronger overall detection capability in building change scenarios.

On the WHU-CD dataset, DPGNet shows even clearer advantages, achieving an $F1$-score of 94.49%, an IoU of 89.55%, and an OA of 99.57%, which are 0.36%, 0.64%, and 0.06% higher than those of the second-best method, BiFA, respectively. Meanwhile, its precision and recall reach 95.49% and 93.51%, maintaining a good balance between the two and demonstrating stronger adaptability in practical application scenarios.

On the SYSU-CD dataset, although the scenes are more complex and the change types are more diverse, DPGNet still achieves the best $F1$-score of 84.16% and IoU of 72.64%. Despite not achieving the highest precision and recall simultaneously, the leading overall metrics indicate that DPGNet achieves a better balance between them and stronger overall discrimination in complex scenes.

In summary, the quantitative comparison results on the three datasets fully demonstrate the effectiveness and superiority of DPGNet. This mainly benefits from its progressive difference-prior modeling capability, as well as the closed-loop collaborative optimization mechanism that links prior generation, retrospective verification, and guided aggregation.

2) Qualitative Analysis: To further illustrate the detection performance of the proposed method, qualitative analysis is conducted on the LEVIR-CD, WHU-CD, and SYSU-CD datasets (see Figs. 7–9). In the visual results, FPs are shown in red, FNs in green, TPs in white, and TNs in black. The visual results on typical difficult samples show that the proposed method has clear advantages in the following aspects.

a) Advantage in pseudo-change suppression: In CD, illumination, shadows, and seasonal variation can easily induce pseudo changes, which in turn lead to false alarms. As shown in Figs. 7(d) and 9(f), under strong illumination changes, some methods mistake brightness variations on building surfaces

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

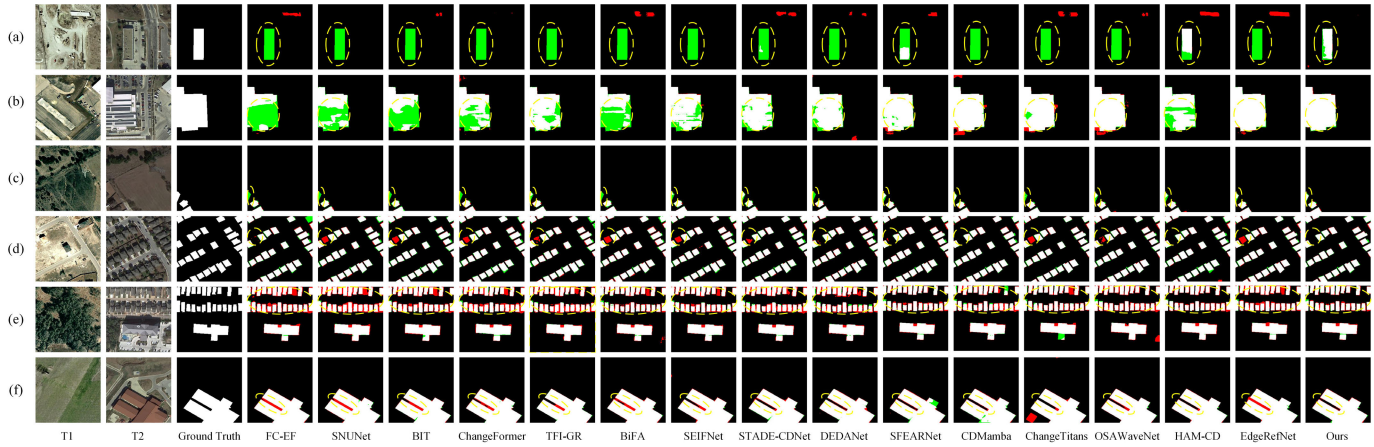


Fig. 7. Visual comparison results of different methods on the LEVIR-CD dataset, where (a)–(f) denote six typical samples.

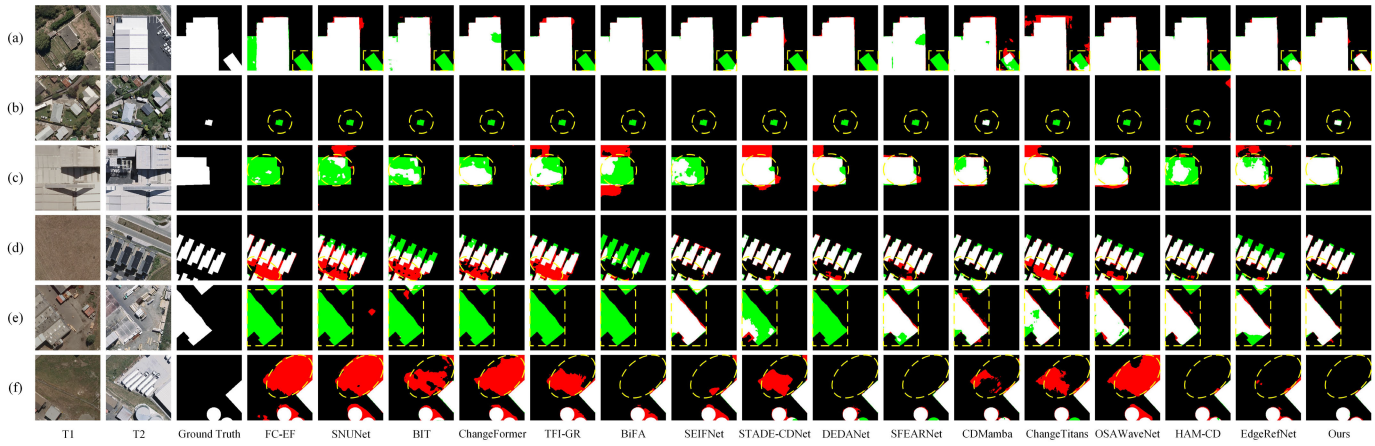


Fig. 8. Visual comparison results of different methods on the WHU-CD dataset, where (a)–(f) denote six typical samples.

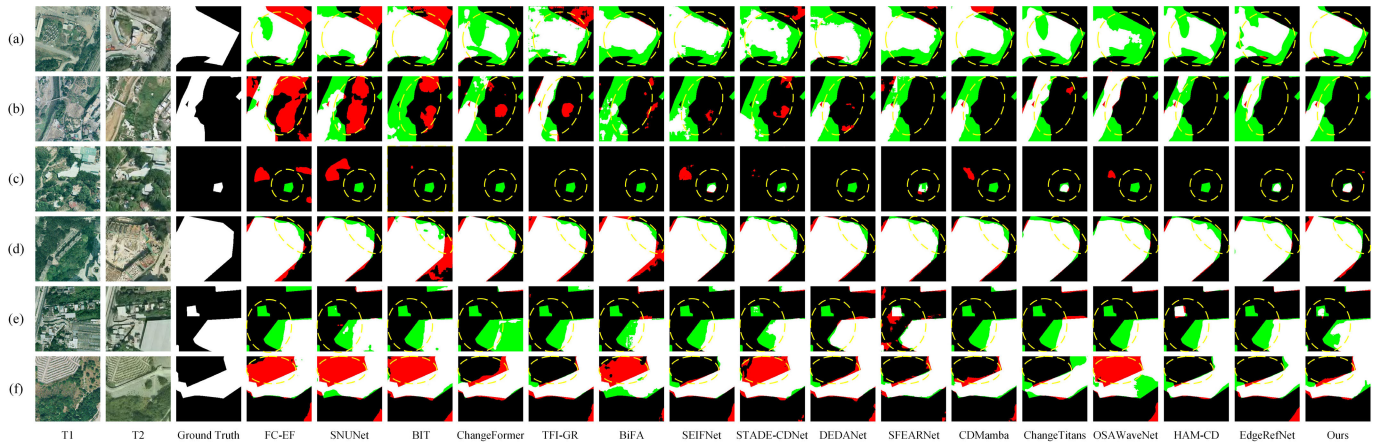


Fig. 9. Visual comparison results of different methods on the SYSU-CD dataset, where (a)–(f) denote six typical samples.

for real changes, whereas DPGNet better suppresses such interference. Similarly, in the shadowed scenes of Figs. 7(f) and 8(d), existing methods tend to misclassify shadow regions as changed targets, while DPGNet more accurately distinguishes shadow variations from real building changes. In Fig. 9(b), where seasonal changes increase vegetation, DPGNet also reduces irrelevant background activations. These results demonstrate its stronger suppression capability in typical pseudo-change scenarios.

b) Advantage in multiscale target perception: A key challenge in CD is to simultaneously ensure stable detection of large-scale changes and effective identification of small-scale changes within a unified framework. As shown in Figs. 7(a) and 8(e), for large buildings with backgrounds

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

TABLE II

MODEL EFFICIENCY COMPARISON OF DIFFERENT METHODS WITH BITEMPORAL INPUT IMAGES OF SIZE 256 × 256

Method	Params (M)	FLOPs (G)	Time (ms)	FPS	Memory (MB)
FC-EF	1.35	3.58	2.15	465.86	61.52
SNUNet	1.35	4.73	2.90	342.22	67.19
BIT	12.40	10.63	9.36	106.83	84.08
ChangeFormer	41.03	202.79	24.10	41.49	461.26
TFI-GR	28.37	9.74	6.50	153.80	144.15
BiFA	9.88	53.00	28.81	34.71	751.55
SEIFNet	28.60	8.26	7.16	139.61	155.14
STADE-CDNet	12.40	11.98	18.97	52.72	199.51
DEDANet	105.14	12.14	70.84	14.12	695.29
SFEARNet	5.56	4.65	19.12	52.30	99.98
CDMamba	6.84	49.68	35.59	28.10	282.47
ChangeTitans	27.15	30.02	188.28	5.31	326.80
OSAWaveNet	27.86	7.68	21.75	45.97	149.15
HAM-CD	49.42	22.21	66.38	15.06	252.99
EdgeRefNet	46.29	35.49	28.67	34.88	337.24
DPGNet(Ours)	7.75	8.66	18.50	54.05	112.00

TABLE III

ABLATION RESULTS OF THE PROPOSED DPGNET ON THE LEVIR-CD DATASET

SPRM	DRP	SPGA	Pre	Rec	F1	IoU	OA
✗	✓	✓	91.10	91.60	91.35	84.08	99.12
✓	✗	✓	91.35	92.05	91.70	84.67	99.15
✓	✓	✗	93.12	90.25	91.66	84.61	99.16
✓	✓	✓	92.50	92.20	92.35	85.80	99.22

All evaluation results are expressed as percentages (%), with the best values shown in red.

TABLE IV

ABLATION RESULTS OF THE PROPOSED DPGNET ON THE WHU-CD DATASET

SPRM	DRP	SPGA	Pre	Rec	F1	IoU	OA
✗	✓	✓	94.20	93.00	93.60	87.96	99.50
✓	✗	✓	94.05	93.50	93.77	88.27	99.52
✓	✓	✗	96.15	91.80	93.92	88.55	99.53
✓	✓	✓	95.49	93.51	94.49	89.55	99.57

All evaluation results are expressed as percentages (%), with the best values shown in red.

TABLE V

ABLATION RESULTS OF THE PROPOSED DPGNET ON THE SYSU-CD DATASET

SPRM	DRP	SPGA	Pre	Rec	F1	IoU	OA
✗	✓	✓	85.95	80.65	83.22	71.26	92.33
✓	✗	✓	86.25	81.05	83.57	71.78	92.49
✓	✓	✗	86.65	81.35	83.92	72.29	92.65
✓	✓	✓	86.90	81.59	84.16	72.64	92.76

All evaluation results are expressed as percentages (%), with the best values shown in red.

of similar appearance, some methods suffer from different degrees of missed detection, whereas DPGNet still accurately identifies the main changed regions. Meanwhile, in the small-scale building change scenes shown in Figs. 7(c) and 8(b), DPGNet also maintains good detection performance. These observations indicate that the proposed method has strong adaptability to multiscale change targets.

c) Advantage in structural detail preservation: In complex building change scenes, existing methods often suffer from internal structural fragmentation and blurred boundaries. As shown in Figs. 7(b) and 8(c), although some methods can detect the main changed regions, they still exhibit internal discontinuities or local missing parts, whereas DPGNet better preserves the structural integrity within targets. In the blurred-boundary scene of Fig. 9(d), other methods generally show boundary shifts or contour ambiguity, while DPGNet can still more accurately recover the boundaries of changed regions. These results indicate that DPGNet has a stronger ability to preserve structural details.

3) Model Efficiency: In addition to quantitative and qualitative analysis, we further evaluate the efficiency of different methods from five aspects: the number of parameters (Params), floating-point operations (FLOPs), inference time for a single bitemporal image pair (time), frames per second (FPS), and peak GPU memory consumption during inference (memory). In the efficiency comparison, the input image size is uniformly set to 256 × 256. The detailed results are shown in Table II.

Benefiting from the lightweight MobileNetV2 backbone and the efficient overall design, DPGNet shows clear advantages in computational efficiency. As shown in Table II, DPGNet contains only 7.75-M parameters and 8.66-G FLOPs, which are considerably lower than those of complex models such as ChangeFormer, BiFA, HAM-CD, and EdgeRefNet. Meanwhile, DPGNet requires a peak GPU memory consumption of 112 MB, which is lower than that of most comparison methods. Together with an inference time of 18.50 ms and an FPS of 54.05, these results indicate that DPGNet achieves a favorable balance among model size, computational complexity, inference speed, and memory consumption.

In addition, Fig. 10 further illustrates the overall performance of different methods from the perspectives of both accuracy and efficiency. Compared with the high-complexity DEDANet, DPGNet achieves gains of 1.11% in $F1$-score and 1.91% in IoU on the LEVIR-CD dataset using only about 1/14 of the parameters and lower computational cost. Compared with the relatively lightweight BIT, DPGNet still improves the $F1$-score and IoU on the SYSU-CD dataset by 7.14% and 10.01%, respectively. Meanwhile, compared with recent methods, HAM-CD and EdgeRefNet, DPGNet still achieves higher accuracy while maintaining a good balance among parameters, FLOPs, and inference speed. These results fully demonstrate that DPGNet can achieve high-performance detection under high efficiency, highlighting its practical value.

D. Ablation Study

1) Effectiveness Analysis of Each Module: To evaluate the contribution of each component in DPGNet, module-level ablation experiments are conducted on the LEVIR-CD, WHU-CD, and SYSU-CD datasets (see Tables III–V). For SPRM ablation, the initial difference representations obtained by concatenation and subtraction are retained, while its two core components, namely, the CPF branch and the DPS branch,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

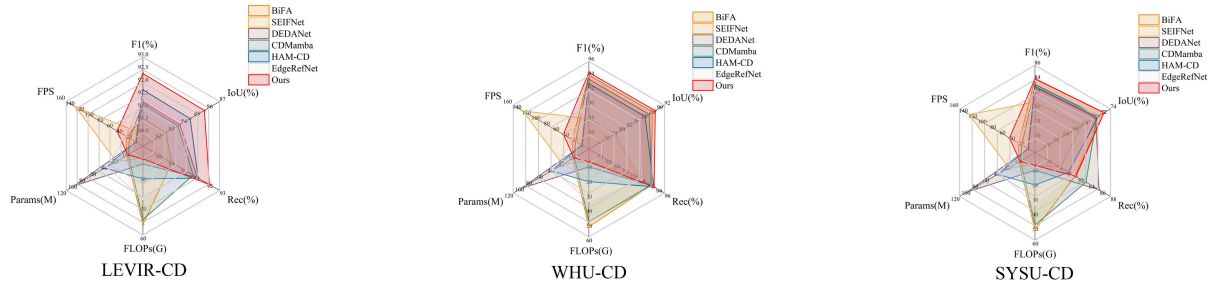


Fig. 10. Radar chart of the overall performance and efficiency of different methods on the LEVIR-CD, WHU-CD, and SYSU-CD datasets.

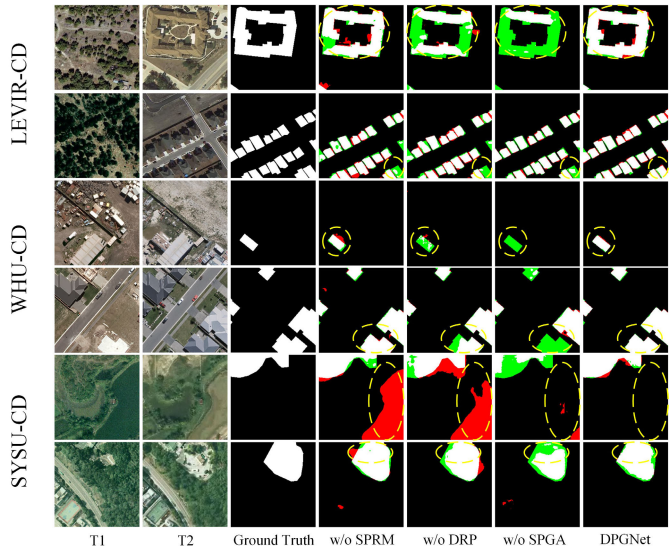


Fig. 11. Visual comparison results under different ablation settings of DPGNet modules. Red, green, white, and black denote FPs, FNs, TPs, and TNs, respectively.

are removed. For DRP ablation, the entire module is directly removed. For SPGA ablation, the basic fusion and output path of adjacent-scale features in the decoding stage are retained, while its intralevel enhancement and semantic prior-guided aggregation mechanisms are removed.

As shown in Tables III–V, the complete model consistently achieves better overall performance than all ablation variants across the three datasets. After removing SPRM, the precision drops to the lowest level on all three datasets, decreasing from 92.50% to 91.10% on LEVIR-CD. After removing DRP, all overall metrics decline. After removing SPGA, the recall drops to 90.25%, 91.80%, and 81.35% on LEVIR-CD, WHU-CD, and SYSU-CD, respectively, all lower than those of the complete model. Especially on the more complex SYSU-CD dataset, the complete model achieves the best results on all five metrics, further demonstrating the effectiveness of the proposed framework.

Fig. 11 presents the visual ablation results of different DPGNet modules. After removing SPRM, obvious false-positive regions reappear in multiple samples, especially in scenes with strong illumination and seasonal interference, indicating that this module is crucial for pseudo-change

TABLE VI
ABLATION RESULTS OF DIFFERENT STATISTICAL PRIOR CONFIGURATIONS IN SPRM

P_{com}	P_{diff}	LEVIR-CD		WHU-CD		SYSU-CD	
		F1	IoU	F1	IoU	F1	IoU
✗	✗	91.46	84.26	93.71	88.16	83.34	71.44
✗	✓	91.63	84.55	93.88	88.47	83.47	71.63
✓	✗	92.08	85.32	94.11	88.87	83.91	72.28
✓	✓	92.35	85.80	94.49	89.55	84.16	72.64

All evaluation results are expressed as percentages (%), with the best values shown in red.

suppression. After removing DRP, unreliable responses in complex regions are more likely to be retained, and real targets with blurred boundaries or weak changes become harder to recover, leading to increases in both false detections and missed detections. After removing SPGA, the detection completeness of targets at different scales declines noticeably, as reflected by insufficient coverage of large buildings and more missed detections of small targets. In contrast, the complete model more stably suppresses interference and preserves complete detection of change targets at different scales.

2) Effectiveness Analysis of Statistical Prior Design in SPRM: To verify the necessity of the two statistical priors designed in SPRM, we conduct ablation experiments on all three datasets, and the results are reported in Table VI. As can be seen, removing both P_{com} and P_{diff} leads to clear performance drops on all three datasets. Specifically, the $F1$-scores decrease from 92.35%, 94.49%, and 84.16% to 91.46%, 93.71%, and 83.34% on LEVIR-CD, WHU-CD, and SYSU-CD, respectively. This indicates that the original difference representations alone are insufficient to effectively suppress interference caused by imaging-style variations and complex backgrounds. Retaining only one prior partially restores performance, but it still remains inferior to the full configuration. Specifically, keeping only P_{com} performs slightly better overall than keeping only P_{diff}, suggesting that the commonality statistical prior plays a more direct role in stabilizing shared background distributions across bitemporal inputs and alleviating pseudo-change interference, but has limited capability in enhancing difference responses in real change regions. In contrast, the discrepancy statistical prior places more emphasis on strengthening cross-temporal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

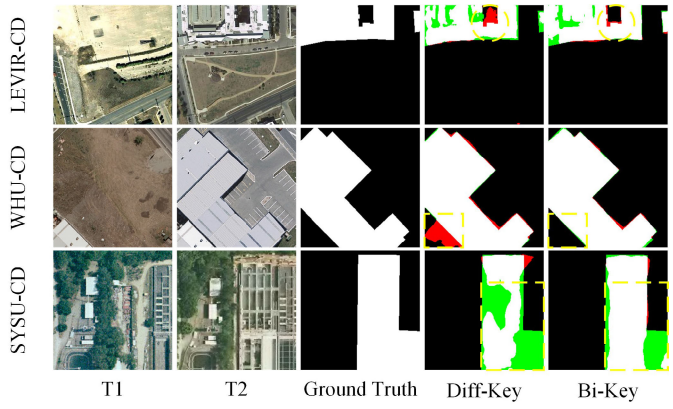


Fig. 12. Visual comparison of different retrospective evidence sources in DRP. Diff-Key and Bi-Key denote the difference-based key and bitemporal-based key settings. Red, green, white, and black denote FPs, FNs, TPs, and TNs, respectively.

TABLE VII
COMPARISON RESULTS OF DIFFERENT KEY-SIDE EVIDENCE SOURCES IN DRP

Key source	LEVIR-CD		WHU-CD		SYSU-CD	
	F1	IoU	F1	IoU	F1	IoU
Difference-based Key	91.96	85.13	94.08	88.81	83.98	72.39
bitemporal-based Key	92.35	85.80	94.49	89.55	84.16	72.64

All evaluation results are expressed as percentages (%), with the best values shown in red.

discrepancy responses and thus provides supplementary cues for true change regions. The full configuration achieves the best results on all three datasets, demonstrating that the two priors play complementary roles in SPRM and that their joint use yields more reliable difference representations.

3) Comparison of Evidence Construction Strategies in DRP: To analyze the effect of evidence construction strategies on different retrospective verification in DRP, we compare two key-generation designs: one uses difference features to generate keys, and the other uses bitemporal features. In both settings, queries are always generated from different features, and values are always generated from bitemporal features. As shown in Table VII, generating keys from bitemporal features yields better performance on all three datasets. In DRP, the initial difference is treated as a change prior to being verified, and its reliability should be determined with the assistance of the original bitemporal context. If the keys are also derived from different features, the matching process tends to remain within the different representations themselves, lacking independent verification evidence from the original temporal features, which makes pseudo responses difficult to distinguish effectively. In contrast, using bitemporal features as keys allows different queries to retrospectively retrieve supporting evidence from the original temporal context, so that real changes can be confirmed and pseudo responses can be suppressed. This is more consistent with the design goal of DRP: treating the initial difference as a prior to be verified through retrospective validation. Fig. 12 further shows that using bitemporal features as keys alleviates the residual

TABLE VIII
ABLATION RESULTS OF DIFFERENT SEMANTIC PRIOR CONSTRUCTION STRATEGIES IN SPGA

Variant	LEVIR-CD		WHU-CD		SYSU-CD	
	F1	IoU	F1	IoU	F1	IoU
w/o Semantic Prior	91.82	84.88	94.20	89.04	83.97	72.37
Semantic Prior (D_f^4 only)	92.10	85.36	94.36	89.32	84.12	72.59
Full SPGA	92.35	85.80	94.49	89.55	84.16	72.64

All evaluation results are expressed as percentages (%), with the best values shown in red.

pseudo responses and missed detections caused by difference-feature-based keys, producing results closer to the ground truth and validating the effectiveness of bitemporal context for retrospective verification.

4) Effectiveness Analysis of Semantic Prior Construction in SPGA: To verify the necessity of semantic prior construction in SPGA, we conduct an ablation study on its semantic-prior guidance mechanism, with the results shown in Table VIII. Specifically, "w/o semantic prior" removes semantic-prior guidance during aggregation, while "semantic prior (D_f^4 only)" constructs the semantic prior only from the biggest difference feature D_f^4. As shown in Table VIII, removing the semantic prior leads to performance drops on all three datasets; using only D_f^4 improves the performance, but still underperforms the full SPGA. These results indicate that semantic priors help alleviate the inconsistency between shallow details and deep semantics, and that the regional structural information in D_f^3 is complementary to the high-level semantic information in D_f^4. Their joint use can more effectively guide shallow feature aggregation.

5) Comparative Analysis of Different Difference Feature Generation Strategies: To further verify the rationality of SPRM design in the difference feature generation stage, we conduct quantitative comparisons with two commonly used basic operations, subtraction and concatenation, as well as two recently proposed methods, ST-DEMs [29] and BDA [41]. ST-DEMs adopts a dual-branch architecture: one branch models difference features using coordinate attention, while the other processes concatenated features through convolution. BDA jointly models subtraction- and concatenation-based features through multibranch cross fusion, combining cross-branch relation modeling based on strip convolutions with a visual center mechanism. As shown in Table IX, SPRM achieves the best results on both LEVIR-CD and SYSU-CD. On WHU-CD, although the precision of SPRM is slightly lower than that of BDA, it still performs best on the other metrics, indicating that SPRM can generate more discriminative difference features by jointly modeling shared background consistency and differential change responses.

The visual heatmaps of different feature generation strategies are shown in Fig. 13. Concatenation is most prone to producing large nontarget high responses under complex backgrounds. As shown in the third and sixth rows, unchanged regions are clearly activated. Although subtraction can highlight the main change regions, noticeable background false responses can still be observed, and the responses within some

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

TABLE IX

QUANTITATIVE COMPARISON OF THE PROPOSED SPRM WITH OTHER DIFFERENCE FEATURE GENERATION STRATEGIES

Dataset	Diff_Gene	Pre	Rec	F1	IoU
LEVIR-CD	Concatenation	90.85	91.65	91.25	83.91
	Subtraction	91.95	90.90	91.42	84.19
	ST-DEMs	92.10	91.35	91.72	84.71
	BDA	92.35	91.80	92.07	85.31
	SPRM	92.50	92.20	92.35	85.80
WHU-CD	Concatenation	93.50	92.80	93.15	87.18
	Subtraction	94.45	92.65	93.54	87.87
	ST-DEMs	94.60	93.20	93.89	88.49
	BDA	95.85	92.20	93.99	88.66
	SPRM	95.49	93.51	94.49	89.55
SYSU-CD	Concatenation	85.95	80.10	82.92	70.82
	Subtraction	85.50	80.85	83.11	71.10
	ST-DEMs	86.45	81.35	83.82	72.15
	BDA	86.85	80.85	83.75	72.04
	SPRM	86.90	81.59	84.16	72.64

All evaluation results are expressed as percentages (%), with the best values shown in red.

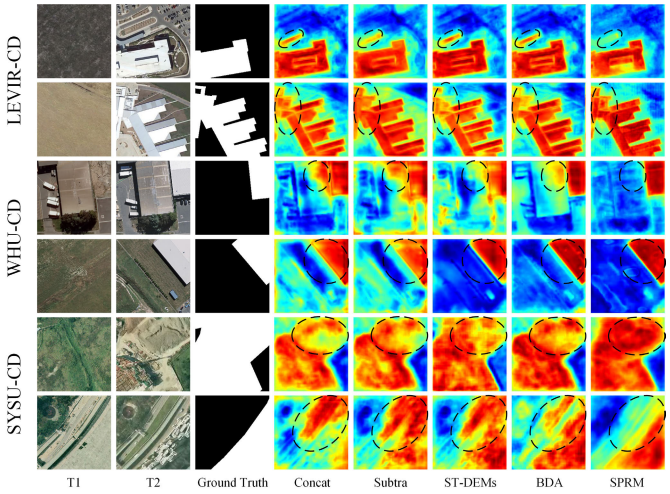


Fig. 13. Visual heatmaps of different feature generation strategies on the three datasets.

targets remain discontinuous. This indicates that, while explicit differencing helps enhance change contrast, it is still difficult to simultaneously suppress pseudo changes and preserve regional integrity. ST-DEMs and BDA can suppress irrelevant activations to some extent, but local pseudo-change residues still appear in the first and sixth rows, and the responses near boundaries remain insufficiently concentrated, resulting in unclear delineation of change-region contours. In contrast, SPRM produces weaker activations on nontarget regions while preserving the internal structures and boundary details of change regions more completely. Both quantitative and qualitative results demonstrate that SPRM can make fuller use of the statistical information in bitemporal features to generate more discriminative and robust difference representations.

TABLE X

QUANTITATIVE COMPARISON OF THE PROPOSED SPGA WITH OTHER DIFFERENCE FEATURE FUSION METHODS

Dataset	Diff_Fuse	Pre	Rec	F1	IoU
LEVIR-CD	FPN	91.55	91.30	91.42	84.20
	ACFM	91.92	91.65	91.78	84.81
	HADF	92.20	91.95	92.07	85.31
	SPGA	92.50	92.20	92.35	85.80
WHU-CD	FPN	94.35	92.60	93.47	87.73
	ACFM	94.80	92.95	93.87	88.44
	HADF	95.10	93.20	94.14	88.93
	SPGA	95.49	93.51	94.49	89.55
SYSU-CD	FPN	85.20	82.25	83.70	71.97
	ACFM	86.15	81.10	83.55	71.75
	HADF	87.55	80.45	83.85	72.19
	SPGA	86.90	81.59	84.16	72.64

All evaluation results are expressed as percentages (%), with the best values shown in red.

6) Comparative Analysis of Different Difference Feature Fusion Strategies: To further verify the rationality of SPGA design in the multiscale difference feature fusion stage, we compare it with three representative fusion strategies, including FPN, ACFM [29], and HADF [40]. FPN performs multiscale aggregation by progressively upsampling and fusing adjacent-level features. ACFM upsamples high-level difference features to align them with low-level features, and then adaptively fuses them by computing fusion weights. HADF first aligns deep and shallow difference features, then models interlevel complementarity through channel-interaction attention and spatial-interaction attention, and finally combines the two attention results to produce the output feature. As shown in Table X, SPGA achieves better overall performance on all three datasets. In particular, it outperforms the other fusion strategies overall on LEVIR-CD and WHU-CD. On the more complex SYSU-CD dataset, although its precision is not the highest, its $F1$-score and IoU still rank first. These results indicate that explicitly using deep difference semantics as priors can more effectively improve the aggregation quality of multiscale difference features.

The visual heatmaps of different feature fusion strategies are shown in Fig. 14. For the small change targets in the first and fourth rows, FPN tends to produce weak responses or even miss detections. Although ACFM and HADF show some improvement, their responses are still not sufficiently concentrated. In contrast, SPGA produces clearer and more complete target responses. For the large-scale change regions in the sixth row, FPN, ACFM, and HADF still suffer from internal discontinuities and incomplete boundaries, whereas SPGA yields more complete and uniform responses. These results indicate that semantic prior-guided aggregation is more effective in balancing small-target perception and complete detection of large-scale targets.

7) Effectiveness Analysis of the Loss Function: To verify the effectiveness and parameter stability of the hybrid loss, we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

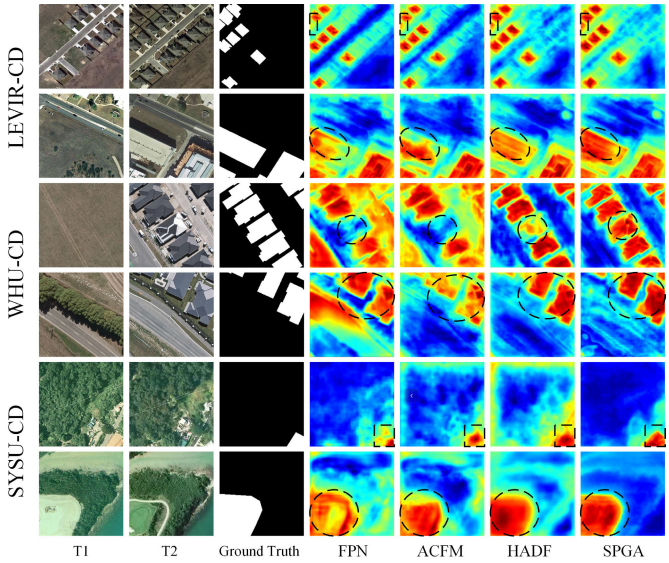


Fig. 14. Visual heatmaps of different feature fusion strategies on the three datasets.

TABLE XI
ABLATION RESULTS OF DIFFERENT LOSS FUNCTIONS ON THE THREE DATASETS

Loss function	LEVIR-CD		WHU-CD		SYSU-CD	
	F1	IoU	F1	IoU	F1	IoU
Focal	91.68	84.64	93.63	88.02	83.53	71.73
Dice	92.09	85.34	94.27	89.17	83.87	72.24
Dice+Focal	92.35	85.80	94.49	89.55	84.16	72.64

All evaluation results are expressed as percentages (%), with the best values shown in red.

TABLE XII
SENSITIVITY ANALYSIS RESULTS OF HYPERPARAMETERS IN THE HYBRID LOSS FUNCTION

α	γ	λ	LEVIR-CD		WHU-CD		SYSU-CD	
			F1	IoU	F1	IoU	F1	IoU
0.25	4	0.1	92.03	85.25	94.20	89.05	83.70	71.98
0.25	4	0.3	92.29	85.70	94.42	89.43	84.01	72.41
0.25	4	0.5	92.35	85.80	94.49	89.55	84.16	72.64
0.25	4	0.7	92.18	85.51	94.17	89.00	84.09	72.54
0.25	4	1.0	91.96	85.12	93.88	88.48	83.83	72.18
0.25	2	0.5	92.16	85.46	94.33	89.27	83.92	72.29
0.25	6	0.5	92.08	85.32	94.05	88.77	83.75	72.04
0.10	4	0.5	92.14	85.43	94.12	88.90	83.88	72.25
0.50	4	0.5	92.21	85.55	94.26	89.14	84.03	72.46

All evaluation results are expressed as percentages (%), with the best values shown in red.

conduct loss-term ablation and parameter sensitivity analyses. First, dice loss, focal loss, and dice + focal loss are used for training, and the results are reported in Table XI. Dice + focal loss achieves the best F1 and IoU on LEVIR-CD, WHU-CD, and SYSU-CD, indicating that the two losses are complementary and provide more effective optimization constraints.

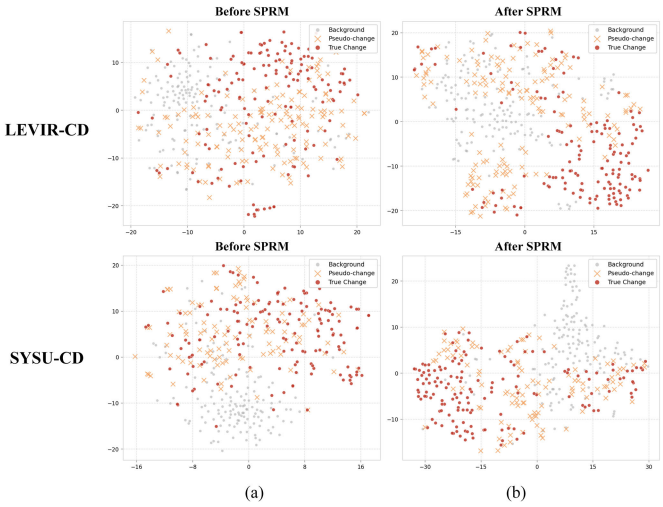


Fig. 15. t-SNE visualization of the effect of SPRM on feature distributions. (a) Distribution of initial difference features obtained by elementwise subtraction of bitemporal features. (b) Feature distribution after SPRM processing. Category-balanced sampling is adopted in the figure, so the number of samples does not reflect the true class proportions.

Second, a one-factor-at-a-time sensitivity analysis is performed on α and γ in focal loss and the hybrid loss weight λ, as shown in Table XII. The model achieves the best performance on all three datasets when $\alpha = 0.25$, $\gamma = 4$, and $\lambda = 0.5$. The small performance variations under different settings indicate good stability, while excessively small or large λ and overly large γ cause performance degradation, further validating the default settings.

V. DISCUSSION

A. Visualization Analysis of Feature Distributions in SPRM

To verify the style-calibration ability of the proposed module on different features, t-SNE is adopted to visualize the distribution of different representations. Three types of samples are selected for analysis: background, pseudo changes, and real changes. As shown in Fig. 15, before calibration, pseudo changes are noticeably mixed with real changes, and some pseudo changes deviate from the background distribution. After processing, the distribution of the three categories becomes clearer: real-change samples are more compact, while pseudo-change samples are more concentrated in the transitional region between categories. These results indicate that the proposed module can alleviate interference from imaging variations, such as illumination and seasonal changes, while improving the discriminability of difference representations.

B. Limitations and Future Work

Although DPGNet achieves a favorable balance between accuracy and efficiency in CD, it still has limitations in a few extreme scenarios. As shown in Fig. 16, when change targets are extremely small and highly scattered, their local discriminative information and contextual support are limited, and DPGNet may still suffer from local missed detections or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

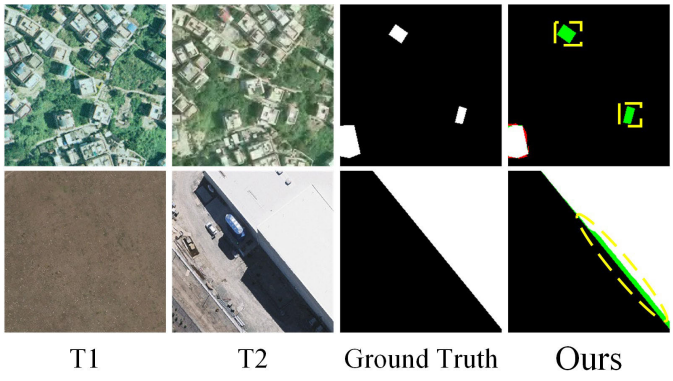


Fig. 16. Failure cases of DPGNet in challenging scenes. Red, green, white, and black denote FPs, FNs, TPs, and TNs, respectively.

incomplete predictions. Future work may explore finer-grained change representations to better characterize small, scattered change targets with low proportions and weak responses. In addition, for building change regions with ambiguous boundaries, especially when changed regions are adjacent to shadows or have low local boundary contrast, the model may still incompletely identify edge changes. Boundary constraints or structural consistency modeling can be introduced in future work to better characterize ambiguous building boundaries.

VI. Conclusion

In this article, we propose DPGNet for the RSCD task. Centered on active difference-prior guidance, the network builds a closed-loop optimization framework for difference representation calibration, retrospective verification, and guided aggregation. Specifically, DPGNet uses SPRM to actively calibrate initial discrepancies, employs DRP to retrospectively query bitemporal context for change authenticity verification, and adopts SPGA for semantic-guided aggregation of multiscale difference features. Experimental results on three benchmark datasets show that DPGNet outperforms SOTA methods in detection accuracy while remaining computationally competitive. In addition, as a convolutional architecture that does not rely on Transformers or state-space modules, it is more hardware-friendly and easier to deploy on resource-constrained platforms such as FPGAs.

Nevertheless, DPGNet still has room for improvement in a few extreme cases, such as extremely small and highly scattered change targets and building change regions with ambiguous boundaries. In future work, we will further explore finer-grained change representation and boundary structural consistency modeling to improve the detection capability in these challenging cases. Moreover, we will further explore the active difference-prior guidance mechanism in weakly supervised CD. Since weakly supervised settings lack pixel-level supervision, change discrimination relies more heavily on effective prior constraints and supporting evidence. Therefore, the difference-prior-guided framework proposed in this article is expected to provide a natural foundation for CD under limited supervision.

References

[1] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022.

[2] S. Dong, Y. Zhu, G. Chen, and X. Meng, "EfficientCD: A new strategy for change detection based with bi-temporal layers exchanged," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639313.

[3] A. H. Chughtai, H. Abbasi, and I. R. Karas, "A review on change detection method and accuracy assessment for land use land cover," *Remote Sens. Appl., Soc. Environ.*, vol. 22, Apr. 2021, Art. no. 100482.

[4] K. Lou, M. Li, F. Li, and X. Zheng, "Integrating local–global structural interaction using Siamese graph neural network for urban land use change detection from VHR satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4413317.

[5] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[6] J. Wang, H. Guo, X. Su, L. Zheng, and Q. Yuan, "PCDASNet: Position-constrained differential attention Siamese network for building damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622318.

[7] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Review ArticleDigital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, May 2004.

[8] P. J. Howarth and G. M. Wickware, "Procedures for change detection using Landsat digital data," *Int. J. Remote Sens.*, vol. 2, no. 3, pp. 277–291, Jul. 1981.

[9] X. Dai and S. Khorram, "Quantification of the impact of misregistration on the accuracy of remotely sensed change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 1997, pp. 1763–1765.

[10] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdiscip. Rev. Comput. Stat.*, vol. 2, no. 4, pp. 433–459, 2010.

[11] T. Han, M. A. Wulder, J. C. White, N. C. Coops, M. F. Alvarez, and C. Butson, "An efficient protocol to process Landsat images for change detection with tasselled cap transformation," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 1, pp. 147–151, Jan. 2007.

[12] D. Seo, Y. Kim, Y. Eo, W. Park, and H. Park, "Generation of radiometric, phenological normalized image based on random forest regression for change detection," *Remote Sens.*, vol. 9, no. 11, p. 1163, Nov. 2017.

[13] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009.

[14] T. Kattenborn, J. Leitloff, F. Schiefer, and S. Hinz, "Review on convolutional neural networks (CNN) in vegetation remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 24–49, Mar. 2021.

[15] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

[16] X. Zhao, L. Wang, Y. Zhang, X. Han, M. Deveci, and M. Parmar, "A review of convolutional neural networks in computer vision," *Artif. Intell. Rev.*, vol. 57, no. 4, p. 99, 2024.

[17] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[18] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[19] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] J. Huang, X. Yuan, C.-T. Lam, W. Ke, and G. Huang, "Large kernel convolution application for land cover change detection of remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104077.

[21] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[22] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[23] Z. Wang, F. Gao, J. Dong, and Q. Du, "Global and local attention-based transformer for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.

[24] J. Bromley, I. Guyon, Y. LeCun, E. Säckinger, and R. Shah, "Signature verification using a 'Siamese time delay neural network," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 6, 1993, pp. 737–744.

[25] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[26] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.

[27] Z. Li et al., "STADE-CDNet: Spatial–temporal attention with difference enhancement-based network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611617.

[28] M. Pan, H. Yang, C. Yu, M. Li, and A. Deng, "A transformer-based multiscale difference enhancement network for change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13778–13793, 2025.

[29] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.

[30] G. Li, P. Han, W. Wang, T. Mu, Z. Xiao, and X. Li, "HAM-CD: Hybrid attention mamba for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5609518.

[31] C. Yang, Z. Wang, Q. Jiang, D. Li, K. Chen, and N. Xu, "Orthogonal semantic axes-guided clustering and frequency-enhanced difference learning for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 10859–10873, 2026.

[32] S. Dong, F. Zuo, G. Chen, S. Fu, and X. Meng, "A remote sensing image change detection method integrating layer-exchange and channel-spatial differences," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 14804–14819, 2025.

[33] S. Fu, S. Dong, and X. Meng, "Beyond cross-temporal difference: Style-aligned and fusion-difference learning for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631415.

[34] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.

[35] J. Zhao et al., "GeoFormer: A geometric representation transformer for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4410617.

[36] D. Lu, S. Cheng, L. Wang, and S. Song, "Multi-scale feature progressive fusion network for remote sensing image change detection," *Sci. Rep.*, vol. 12, no. 1, p. 11968, Jul. 2022.

[37] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[38] Z. Huang et al., "MCECF: A multiscale complementary enhanced context fusion network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4407414.

[39] Y. Xing, J. Hu, Y. Jia, and R. Huang, "Multiscale edge enhancement and progressive change-aware network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 18197–18208, 2025.

[40] L. Wang, Y. Xie, Z. Lu, and S. Tian, "Difference perception fusion network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 16966–16982, 2025.

[41] H. Li, S. Cheng, and A. Du, "Difference enhancement and dependence-aware network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 21384–21399, 2025.

[42] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[43] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[44] Y. Sun, J. Wu, H. Zhang, Z. Li, L. Lei, and G. Kuang, "Iterative global mapping-local searching for heterogeneous change detection with unregistered images," *Int. J. Comput. Vis.*, vol. 134, no. 4, p. 143, Apr. 2026.

[45] H. Wang, N. Wang, and X. Li, "FarmCD: Agricultural information-guided gated network for farmland change detection from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4401714.

[46] Y. Chen, X. Ning, R. Zhang, H. Zhang, X. Huang, and Y. He, "ESMII-Net: An edge-synergy and multidimensional information interaction network for remote sensing change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 139, May 2025, Art. no. 104507.

[47] Y. Liu, S. Li, Z. He, and K. Liu, "Edge and flow guided iterative CNN for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4406422.

[48] W. I. M. Hussin, Z. Lu, A. Ashraf, A. Mao, and Z. Peng, "EdgeRefNet: An edge-guided refinement network for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4409610.

[49] Q. Wang et al., "Spatial–temporal evolution guided change detection network for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 14080–14092, 2024.

[50] G. Xu, Y. Liu, L. Deng, X. Wang, and H. Zhu, "SMNet: A semantic-guided mamba network for remote sensing change detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 5, pp. 11116–11127, Oct. 2025.

[51] K. Song and J. Jiang, "AGCDetNet: An attention-guided network for building change detection in high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4816–4831, 2021.

[52] H. Wei et al., "Multiscale spatial frequency fusion and prior change guidance network for remote sensing change detection," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–13, 2025.

[53] Y. Rao, W. Zhao, Z. Zhu, J. Lu, and J. Zhou, "Global filter networks for image classification," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 980–993.

[54] Y. Liu, F. Zhang, S. Zhang, K. Zhang, J. Sun, and L. Bruzzone, "Content-guided spatial–spectral integration network for change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5604816.

[55] Z. Ying et al., "DGMA2-Net: A difference-guided multiscale aggregation attention network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5619716.

[56] J. Wang et al., "SSCFNet: A spatial–spectral cross fusion network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4000–4012, 2023.

[57] C. Han, C. Wu, H. Guo, M. Hu, J. Li, and H. Chen, "Change guiding network: Incorporating change prior to guide change detection in remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8395–8407, 2023.

[58] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[59] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[60] Z. Li, C. Tang, L. Wang, and A. Y. Zomaya, "Remote sensing change detection via temporal feature interaction and guided refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628711.

[61] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[62] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[63] M. Li, D. Ming, L. Xu, D. Dong, and Y. Zhang, "SFEARNet: A network combining semantic flow and edge-aware refinement for highly efficient remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5402518.

[64] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[65] Z. Yang, G. Pei, Y. Yao, T. Zhou, L. Ding, and F. Shen, "ChangeTitans: Toward remote sensing change detection with neural memory," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4709714.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 12:23:51 UTC from IEEE Xplore. Restrictions apply.